Karlsruhe Institute of Technology

This is the author's version of a work that was published in the following source:

Knierim, M. T., Zimny, C., Seitz, J. (forthcoming). **Monitoring Cognitive Load with 3D-Printed EEG Headphones**. Accepted for Presentation at the 43rd International Conference on Information Systems (ICIS), Copenhagen 2022.

Please note: Copyright is owned by the author and / or the publisher. Commercial use is not allowed.

Institute of Information Systems and Marketing (IISM)
Kaiserstr. 93
76131 Karlsruhe - Germany
https://iism.kit.edu

© 2022. This manuscript version is made available under the CC-BY-NC-ND 4.0 license http://creativecommons.org/licenses/by-nc-nd/4.0/

Monitoring Cognitive Load with 3D-Printed EEG Headphones

Completed Research Paper

Michael Thomas Knierim
Karlsruhe Institute of Technology (KIT)
Kaiserstr. 93, 76131 Karlsruhe, Germany
michael.knierim@kit.edu

Christian Zimny
Karlsruhe Institute of Technology (KIT)
Kaiserstr. 93, 76131 Karlsruhe, Germany
ubelo@student.kit.edu

Julia Seitz
Karlsruhe Institute of Technology (KIT)
Kaiserstr. 93, 76131 Karlsruhe, Germany
julia.seitz@kit.edu

Abstract

Real-world cognitive load monitoring promises to be an essential foundation for positive adaptive information systems that foster knowledge workers' productivity and well-being. Towards this goal, our research combines established EEG load monitoring principles with low-cost, 3D-printed headphones. This provides a means for continuous, unobtrusive, and real-time cognitive load detection. Results from two experiments document strong relationships of cognitive load reports with EEG frequency bands (Theta and Alpha), both around the ears and on top of the head. While limitations with Theta band power sensitivity are observed, Alpha band modulations are robust across sessions, task repetitions, and three tasks. Furthermore, short setup durations are found, and only minimal influences of hairstyle or glasses on setup times and load relationships. With a discussion of the remaining challenges for more naturalistic studies, this article documents the system's future potential for load sensing during knowledge work.

Keywords: Cognitive Load, Wearable EEG, Headphones, OpenBCI, Flow Experience, Open-Source, 3D-Printing

Introduction

Managing mental effort during information system (IS) use in everyday life is an important goal for numerous reasons (Grimes & Valacich, 2015). For instance, related research has most recently demonstrated how cognitive overload can cause detrimental health consequences through the misuse of teleworking-IS during the COVID-19 crisis (Martin et al., 2022). Similarly, earlier works show that general task performance suffers when mental workload increases (Hart & Staveland, 1988) or that knowledge building is impeded by heightened comprehension difficulty (Brünken et al., 2003). Such scenarios can be described as eliciting heightened cognitive load – the mental effort and working memory required to complete a task (Paas et al., 2003). While it is generally not necessary or preferable to eliminate all cognitive load, the ability to monitor disadvantageous load levels has a wide range of practical applications for work, learning, and personal IS use (Grimes & Valacich, 2015; Hu et al., 2017; Lyu et al., 2015). Especially in the expanding domain of knowledge work, the continuous, timely, and unobtrusive monitoring of cognitive load could provide essential benefits for improving productivity, well-being, and growth. For these reasons, scholars are increasingly calling for ubiquitous cognitive load monitoring approaches that could form the foundation of load-regulating adaptive systems and interventions (Grimes & Valacich, 2015; Lyu et al.,

2015). Such systems could support knowledge-creating activities by providing recommendations for breaks, goal adjustments, or interruption reductions (Berka et al., 2007; Rissler et al., 2020).

However, it is still a major challenge to assess cognitive load continuously, unobtrusively, robustly, and with high temporal resolution. While several cognitive load measurement methods exist, there is so far no approach that can be used continuously in real-world situations. Self-reports are easily accessible (see, e.g., Hart & Staveland, 1988) but require constant interruption and can not achieve the required continuity and automated support. Behavioral methods, like the load determination by analyzing mouse or eye-movement dynamics, overcome these problems, yet, are limited by a substantial task dependence (Grimes & Valacich, 2015). Neurophysiological measures (e.g., electroencephalography – EEG) have been found to provide robust and cross-task estimators of cognitive load (Berka et al., 2007) and have even already been employed in load-adaptive IT systems (Blankertz et al., 2016; Ewing et al., 2016). Nevertheless, the major downside of this method is the instrumentation itself – EEG measurements typically rely on gelled electrode caps and long and cumbersome setup procedures with hair-washing and skin preparation (Bleichner & Debener, 2017; Riedl et al., 2020)[1]. Even some of the more portable systems still often come with alienating aesthetic form factors, which make them impractical for real-world settings. Thus, to enable real-world load monitoring, dry-electrode EEG systems are needed that are set up quickly and blend in seamlessly with existing wearable technologies.

For this purpose, we have transformed the OpenBCI open-source, dry-electrode EEG recording system from a standard, helmet-shaped headgear into a more naturally occurring form factor – a pair of 3D-printed headphones. This design allows using established hardware & sensing principles in a low-cost and reproducible wearable device with a commonly accepted form factor. By keeping the open-source nature of all components and by developing a configurable system, we provide a toolkit for scholars with various backgrounds for the study of cognitive load, especially for knowledge work. With this article, we aim to first answer the following research question: *Can cognitive load changes be identified by dry-electrode EEG measurements with sensor locations in the headband and ear regions of typical headphones?*

To answer this question, we conduct two controlled laboratory experiments that assess the systems' general monitoring aptitude across sessions, individuals, and tasks. The results show strong relationships between self-reports of cognitive load and EEG frequency bands, specifically Theta band activity on the top of the head and Alpha band activity both around the ears and on top of the head. While the Theta band relationship shows limitations in terms of reliability and sensitivity, the majority of the observations are highly in line with established EEG findings and document the system's aptitude for load detection. Importantly, towards applying the system in real-world situations, we refrained from any skin or hair preparation of the participants and collected data from people with various hairstyles, genders and glasses. With short setup durations throughout (less than five minutes on average) and only minimal impact on setup times and main effects, our findings provide a solid foundation for extended work.

Thereby, we contribute to the IS discipline by enabling further research on load monitoring in knowledge work scenarios through the provision of hardware, methodology, and measurement validation. All the materials for replicating the described system (list of components and assembly instructions) are made available with the article. While there are limitations to the real-world use of the EEG features as load indices that we cannot fully assess yet (e.g., the influence of movement on electrode contact and signal artefacts or the influence of time and visual stimulation on frequency band powers), we discuss how future work could address these factors. Finally, we highlight an interesting connected use case in the knowledge work context, that is the possibility to observe flow experience occurrence through cognitive load monitoring, which has been discussed as a desirable vehicle for increased productivity and well-being in knowledge work (Knierim et al., 2019; Rissler et al., 2020). Therefore, the theoretical and empirical integration between cognitive load and flow is discussed at the end of the article as a final contribution, together with a proposed study design to assess the system's applicability in a more naturalistic setting.

[1] The cumbersome experience with gelled EEG systems has even caused public criticism from study participants. The following blog post describes a participant's ordeal of finding a cleansing agent to remove EEG gel: http://belovedmonsterandme.blogspot.com/2009/01/glue.html - last accessed 17.08.2022.

Theoretical and Conceptual Background

In general, it can be said that cognitive processing consists of two main components, a rather bounded working memory and a much larger long-term memory (Paas et al. 2003). To process an activity, some level of working memory needs to be devoted to the task at hand. This process is referred to as the exerted level of cognitive load (Grimes & Valacich, 2015; Paas et al. 2003). Cognitive load can thereby be categorized into three types: (1) intrinsic load that is related to a specific topic, (2) extraneous load that is related to how tasks are presented, and (3) germane load related to the work needed to store knowledge permanently (Paas et al. 2003). While this conceptual differentiation is useful for assessing load sources, the three facets are not easily differentiated by measurement methods like EEG. Therefore, we continue this article by considering cognitive load as a holistic construct that encompasses the sum of these three load dimensions.

A substantial amount of research on cognitive load has investigated the detrimental effects of high load on knowledge building in IT-mediated learning scenarios (Brünken et al., 2003; Paas et al., 2003). Beyond this context, an increasing body of work is emerging in the IS discipline and neighboring fields that documents the critical consequences of (prolonged or extensive) cognitive load in the area of knowledge work. For one, several studies are documenting the adverse impact of high load on task performance, caused, for example, by reduced decision speed and accuracy (Grimes & Valacich, 2015) or by extended information search and evaluation processes (Hu et al., 2017). At the same time, these effects are accompanied by reduced IT-system satisfaction (Hu et al., 2017; Lyu et al., 2015) and can even lead to stress and lowered physical and mental well-being (Galy et al., 2012; Lyu et al., 2015). With the predicted shift of work sectors into the knowledge domain due to the extended AI-computerization of various tasks (Frey & Osborne, 2017), we argue that it will become increasingly important to regularly monitor cognitive load in knowledge work.

Several methods can be used to assess cognitive load, including subjective measures (i.e., self-reports), behavioral measures (e.g., learning outcomes), and objective measures like mouse movements, eye-tracking or brain activity (Brünken et al., 2003; Chikhi et al., 2022; Lyu et al., 2015). While each of these approaches provides some insight into the level of experienced cognitive load, they also have shortcomings that researchers and practitioners alike must be aware of – especially when considering which degree of ubiquity and continuity the methods provide for load monitoring. Survey methods suffer from response bias and task interruption, which may lead to confusion, reduction of performance, and dissatisfaction (Grimes & Valacich, 2015). Behavioral methods, like the observation of mouse or eye-movement dynamics are limited by a substantial task dependence (Grimes & Valacich, 2015). For example, in the knowledge work domain, writing code or a scientific article can require minimal mouse activity, and generates specific eye-movements, that are likely very different from, for example, an online-shopping scenario. Neurophysiological measures such as heart rate, pupil dilation, and fMRI can interfere with the natural use of the system, thus inhibiting the user from engaging in their primary task (Grimes & Valacich, 2015). Ubiquitous, real-world load monitoring still needs to overcome the hurdles of high-quality user experiences (comfort, intrusiveness, aesthetics) and high measurement quality (noise and reliability). A promising development in this regard is the recent research on "transparent" EEG (Bleichner & Debener, 2017).

Transparent EEG methods focus on collecting neural signal data through hardware systems that are placed on the head at inconspicuous positions, at the moment primarily in ear-adjacent locations (Bleichner & Debener, 2017). In-ear EEG or around-the-ear EEG electrode arrays are found to be usable for high-quality EEG recordings with much higher convenience than standard EEG cap recordings, for example, in situations like speaker attention detection (Bleichner & Debener, 2017) or affect monitoring (Athavipach et al., 2019). Two articles using around-the-ear cEEGrid electrodes also already demonstrate the possible use of ear EEG for workload monitoring in driving simulators (Wascher et al., 2019) and cognitive tasks (Knierim et al., 2021). These works leverage the established principles of EEG load monitoring. The human EEG has been shown to provide reliable estimators of cognitive load through characteristic modulations in the power of EEG frequency bands (Müller-Putz et al., 2015). The most prominent frequency bands with power changes related to cognitive load are Theta and Alpha. Such frequency band activity is derived by decomposing the EEG signal into oscillatory, wave-like patterns (see Figure 1). In general, Theta waves are characterized by low-frequency activity (4–7 Hz) often associated with various sleep phases, mediation, fatigue, and cognitive load patterns (Müller-Putz et al., 2015). Alpha waves, on the other hand, are characterized by medium-frequency activity (8–12 Hz) and generally indicate states of relaxation and readiness (Berger, 1929).

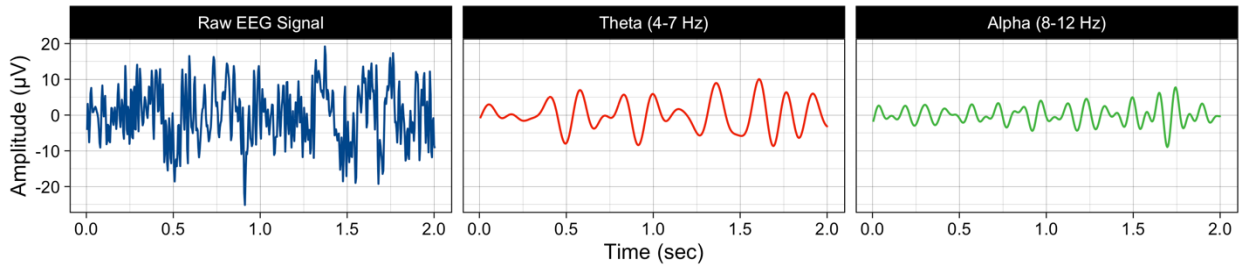

Figure 1. Excerpts of EEG signals with extracted frequency oscillations. The data are from electrode L3 of one study participant during an eyes-closed resting phase.

An increase of this wave type is common and prominent during resting periods with closed eyes. The effect is even so prominent, that it can be seen in the raw EEG signal without signal processing – which is why it was observable in the initial development of the EEG method by Hans Berger in 1929. Thereby, the difference of Alpha power (particularly in posterior brain regions) between eyes-open and eyes-closed rest has been called the Berger effect. Based on this finding, scholars have also argued that Alpha waves represent cognitive inactivity, also referred to as cortical "idling" (Müller-Putz et al., 2015). In relation to cognitive load, Theta power has been shown to be positively correlated with load, most notably in frontal regions, whereas Alpha power is typically found to be negatively correlated with load, particularly in parietal regions (Chikhi et al., 2022). As this pattern has been found to be quite robust, it has been affirmed and used repeatedly in IS literature (for a discussion see Müller-Putz et al., 2015). Scholars have also developed dedicated load indices as a combination of lower and higher frequency powers (e.g., Theta/Alpha frequency band power) that can be used to determine load levels across tasks and situations (Berka et al., 2007). In terms of sensitivity, scholars have even reported being able to predict the exact level in a Tetris game using advanced signal processing methods that isolate the modulation of EEG frequencies related to the cognitive load during that level (Blankertz et al., 2016).

In contrast, the previously mentioned ear-EEG studies (Knierim et al., 2021; Wascher et al., 2019) report limitations in differentiating various cognitive load levels, possibly related to the small amplitudes (resulting from smaller electrode distances). Also, using these cEEGrid electrodes still requires the "typical" steps of recording setups, to achieve high signal qualities (skin preparation, gelled electrodes), and a headband mounting solution that is minimal, but still atypical in real-world situations. For these reasons, we have developed an open-source, 3D printed EEG headphone system that features both greater electrode distances (around the ear and with electrodes on the top of the head – positions that are closer to sites typically used for EEG-based load detection – see, e.g., Ewing et al., 2016; Kartali et al., 2019) and dry electrodes which greatly facilitates recording setups. The following chapters describe the key features of this recording system and the studies we conducted to investigate its usefulness for cognitive load monitoring.

Method & Materials

To develop the dry-electrode EEG headphone system, an available open-source EEG kit and an available open-source headphone project were adapted and joined. This means, that the core of the measurement system is built on established technology, the amplifier and electrodes of the OpenBCI Cyton biosignal acquisition platform. The OpenBCI Cyton amplifier allows 8-channel EEG recordings with a sampling rate of up to 250 Hz, 24 bit resolution, low input-referred noise and low power consumption (Rashid et al., 2018). With a 1000mAh LiPo battery and a 16GB SD card continuous recordings of over 10 hours can be achieved (Knierim et al., 2021). Normally, this amplification system is used with a full head harness, the Ultracortex MK IV (see Figure 2) that features the possibility to use electrode positions of the standard 10-20 reference system (Oostenveld & Praamstra, 2001).

The headphone system, in contrast, allows EEG recordings on three typical electrode positions on the top of the head (C3, Cz, C4), and around the ear, featuring about the same electrode positions that are recorded with the around-the-ear cEEGrid electrodes (Debener et al., 2015) – that we consider as an established reference system for around-the-ear EEG electrode positions. An overview of the available electrode positions in the EEG headphones is shown in Figure 3. Thus, the system's measurement process is also rooted in established methodology. As the Cyton amplifier allows recording from 8 electrodes (plus

reference and ground electrodes), a choice needs to be made for the desired electrode positions in a particular study. In the two following studies, we used position L4 for the reference electrode and position R2 for the ground electrode. This configuration was chosen to allow re-referencing after the recording to a symmetrical linked mastoid reference (L5+R5), which is repeatedly done in cEEGrid studies (Debener et al., 2015; Wascher et al., 2019). As frontal and posterior electrode locations have been found to play an important role for cognitive load monitoring (Chikhi et al., 2022) we used the top electrodes (C3, Cz, C4) and two electrodes at each ear (L3, L5, R3, R4, R5) in the following studies.

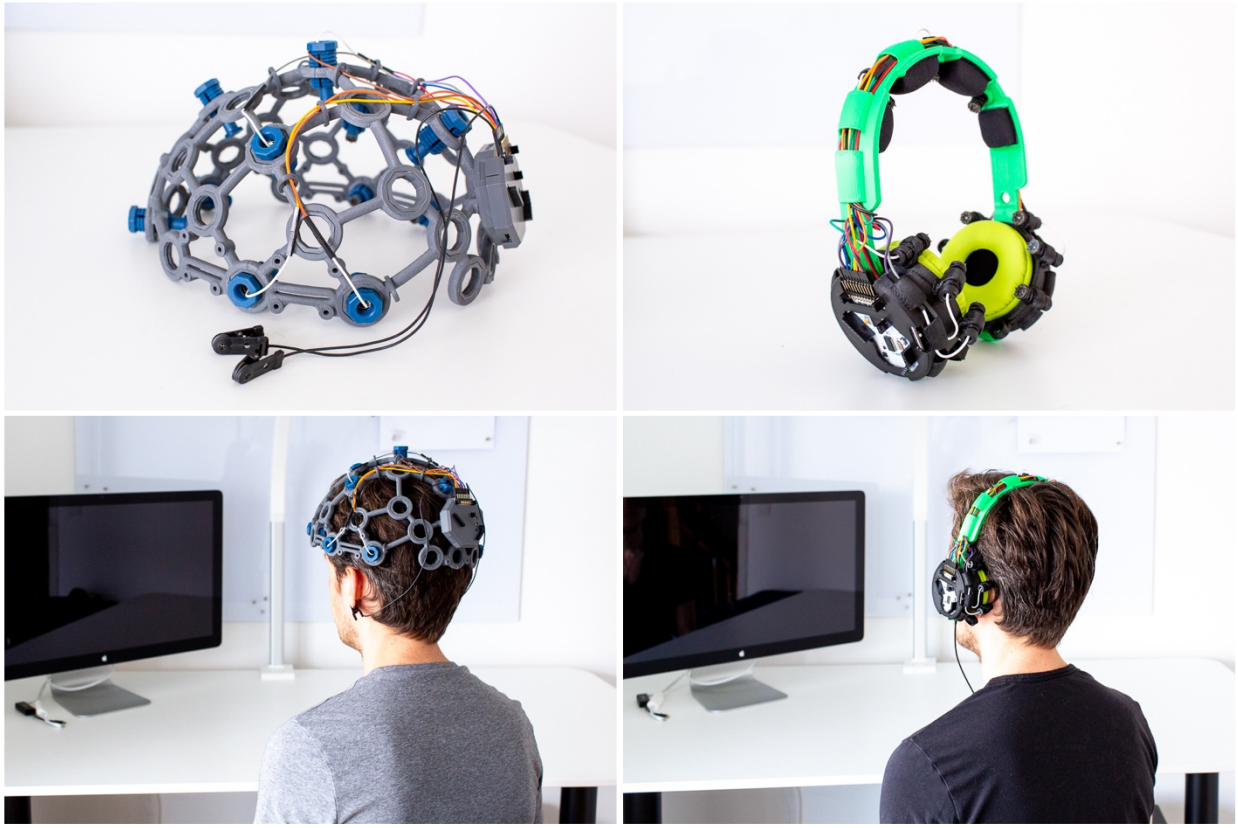

Figure 2. The EEG headphones system (on the right) in comparison to the standard OpenBCI EEG headgear (Ultracortex MK IV) on the left. A similar, but not identical electrode layout configuration is shown.

By building on the knowledge from the "Print.Plus" open-source headphones project, we also leveraged established aesthetic and comfort factors for the headphones design. In addition, the headphones can even be fitted with audio drivers to feature a fully functional headphones system with speakers and a microphone – which can both be used during an EEG recording. All of these components can be freely purchased and assembled at low cost. The latter is especially so because the majority of the components can be generated by regular fused-deposition modeling (FDM) desktop 3D-printers (or through 3D-printing services). The materials for fully replicating the described system (list of components and assembly instructions) are provided here: https://github.com/MKnierim/openbci-headphones.

This headphone EEG solution is not the first of its kind (see Kartali et al., 2019 for a study with a commercial analog with saline water electrodes), yet features a few distinct attributes essential for ubiquitous mental state monitoring research: (1) configurability (selectable electrode positions and open-source modifiability of all hardware and software components), (2) dry electrodes (allowing fast setup), and (3) low cost (facilitating research even with smaller budgets). At the same time, it should be acknowledged, that these advantages also come with their own limitations. Dry electrodes are known to provide poorer signal-to-noise ratios (SNR) than gelled (or saline) electrodes and a higher proneness to electrical and mechanical artefacts (Müller-Putz et al., 2015; Riedl et al., 2020). These limitations should be taken into consideration for various research designs. We propose here though, that the reliance on fast study setups and gel-free

electrodes is likely a key factor in achieving real-world user studies in the long-term. Thus, we argue that our presented system provides a significant bridge into the development of related studies and measurement systems.

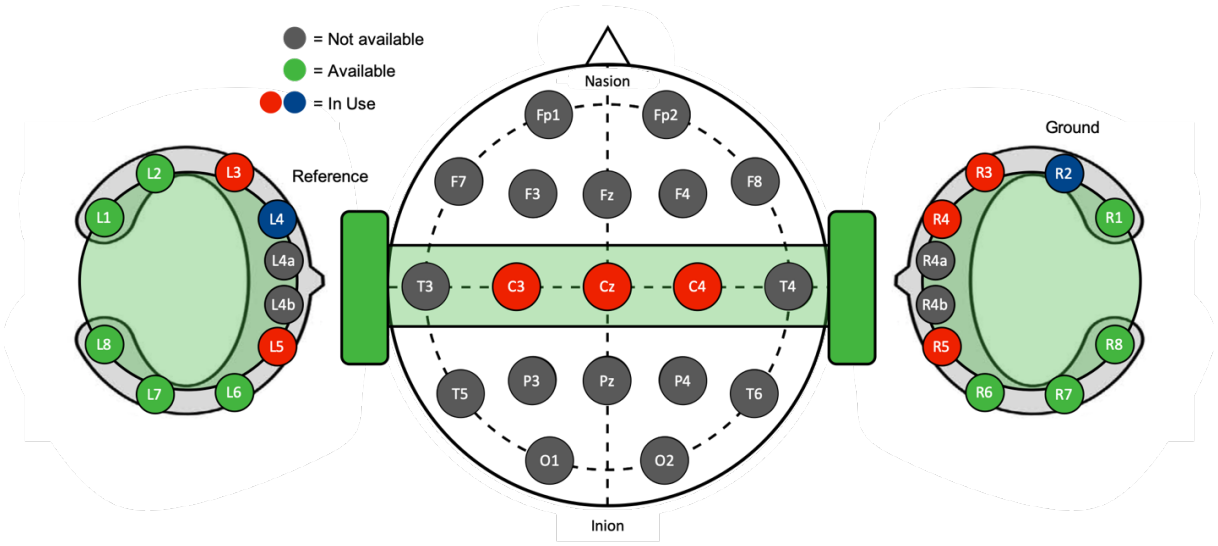

Figure 3. Electrode positions in the EEG headphones system. The configuration for the two studies described below is shown with red and blue electrodes. The layout combines the standard 10-20 electrode positioning system for cap EEG (Oostenveld & Praamstra, 2001) and the layout of around-the-ear cEEGrid electrode arrays as main ear electrode positioning reference (Debener et al., 2015).

To assess the principle measurement capabilities of the system (without going into detailed signal quality aspects – as these have been generally documented for the OpenBCI Cyton amplifier before – see, e.g., Rashid et al., 2018) and to explore the system's suitability for monitoring changes in cognitive load, two laboratory studies were conducted. The details and results of these studies are described in the next two chapters in detail. As, however, the processing of the EEG signals was identical in both studies, we outline these steps here.

The signal processing and feature extraction steps were primarily aligned with the related work on cognitive load observation with around-the-ear cEEGrid electrodes (Knierim et al., 2021; Wascher et al., 2019). This means, that for each 90-seconds data segment (each task round of the experiment), the signals were mean-centered and filtered using a 2-30 Hz finite impulse response (FIR) bandpass filter to remove signal drifts and high-frequency noise. Afterwards, data were epoched (2-second sliding windows with 1-second overlap) and noisy epochs (e.g., due to movement artefacts) were removed as determined by the Riemannian potato similarity criterion using a 2 SD threshold (Barthélemy et al., 2019). Next, the recordings were re-referenced to mean averaged linked mastoids (channels L5+R5). Then, the power spectral density (PSD) was estimated for each epoch using the multitaper method which is suggested when more noisy data can be expected (Babadi & Brown, 2014). All band power estimates were relativized (division of each frequency by the sum of all frequency powers), as this has been found to provide more reliable differentiation for workload monitoring (Wascher et al., 2019). To extract the two frequency bands of interest (Theta and Alpha), the individual Alpha frequency (IAF) was determined visually and confirmed with the "transfreq" clustering method (Vallarino et al., 2021) which detects the transition frequency between the Theta and Alpha band. This step was used to obtain higher spectral precision, as the IAF is known to fluctuate substantially between individuals (Müller-Putz et al., 2015). As the Theta and Alpha bands border each other, a frequency bin of 2 Hz between peak Alpha and peak Theta was kept, reducing spillover effects. Each band was determined as ±1.5 Hz from its determined center. The average Alpha range (based on the average Alpha peak of 10Hz) was 8.5 to 11.5 Hz, and the average Theta range 3.5 to 6.5 Hz.

Study 1

Procedure

To initially assess the potential of the headphones EEG system for neural data collection, a short, controlled experiment was conducted to observe if two main effects can be detected: (1) the Alpha frequency band power increase when eyes are closed (the so-called Berger effect – see, e.g., Debener et al., 2015), and (2) the modulation of Theta and Alpha frequency band powers due to variations in cognitive load. Therefore, the experiment consisted of two task groups: (1) resting phases of eyes-closed and eyes-open rest (with a fixation cross) to study the occurrence of the Berger effect, and (2) a mental arithmetic task in an easier (adding up 3 summands), and a harder condition (adding up 5 summands). Similar procedures have, for example, been used in previous work with around-the-ear cEEGrid studies for sensor validation purposes (Debener et al., 2015; Knierim et al., 2021). To assess changes in cognitive load, the NASA Task Load Index (TLX) was completed after each task round. The TLX is a six-item multi-dimensional self-report instrument that collects perceptions of mental, temporal, and physical load, performance, effort, and frustration (Hart & Staveland, 1988). The TLX has been used in various experiments for similar purposes as ours (see, e.g., Grimes & Valacich, 2015). Figure 4 shows the study procedure with exemplary screenshots from the tasks. To assess the reliability of the main effects, each task was completed three times.

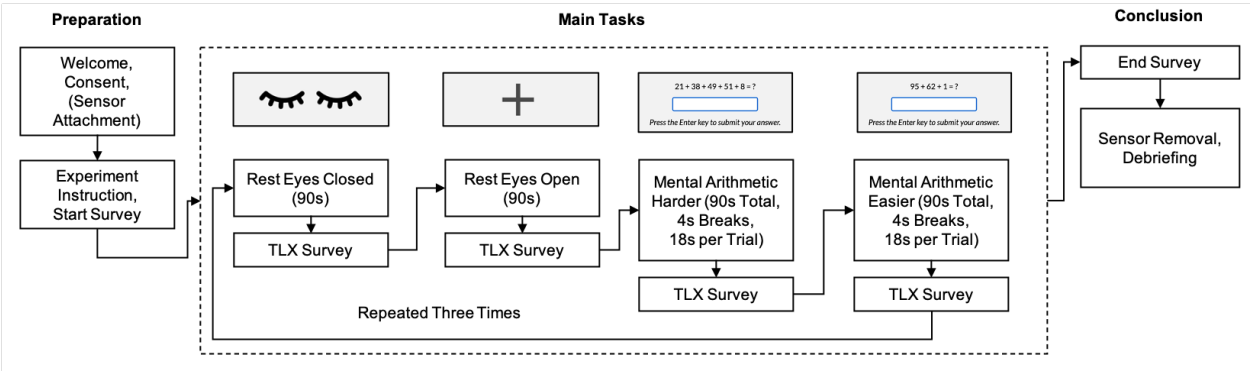

Figure 4. Study 1 procedure overview.

First, a single participant (male, 33 years, wearing glasses – denoted A in Figure 5) completed this experiment four times (once per day) to gage the initial potential of the recording setup and to assess the reliability of detecting the main effects across sessions (within-subject group – WS). This is necessary, as previous EEG research has documented substantial within-subject variance of neural activations across recording sessions (Blankertz et al., 2016; Müller-Putz et al., 2015). Also, as a dry electrode system can be more prone to recording artefacts, it is important to study if an effect can be replicated for a single subject. Afterwards, four additional participants (1 female, age range = 24-31, median age = 28.5, none wearing glasses during the experiment – denoted B-E in Figure 5) completed a single session of the same experiment to also assess the between-subject reliability of the main effects (between subject group – BS). Participants were instructed to limit their movements during the experiment to reduce measurement artefacts.

The average duration of the experiment was 29.11 minutes. All subjects participated voluntarily, without remuneration, and were recruited from the academic staff of the authors' institution. The recordings were conducted in an office building at the authors' university to record data in a traditional office setting, that is, explicitly not a highly controlled laboratory. Upon arrival participants received an introduction to the experiment, the conditions of participation, including an explanation of the instrumentation and the following procedure. Before participation and sensor setup, all participants signed a consent form. Next, the headphone system was set up by the experimentator. That means signals were inspected for low and homogenous amplitudes (between 5 and 15 µV), and absence of PSD outliers which indicate poor electrode impedances on a particular channel. The data was recorded with OpenBCI GUI and the experiment was conducted through a browser interface (LimeSurvey), with local timestamp collection, thereby making sure that a single clock was used for the synchronization of the EEG data and the experiment markers.

Results

Following the recording, the data were processed as outlined above and the main effects investigated. First, the Berger effect was inspected by calculating the Alpha band power difference between eyes closed (EC) and eyes open (EO) resting phases at each electrode and repetition (block). Both the average PSD and the heatmap plot in Figure 5 show a clear and consistent Alpha power increase across sessions, blocks, and electrodes, when the eyes were closed, thereby confirming the ability of the headphone system to robustly detect this known and large neural effect. It should be noted that sporadic variation in the strength of the Alpha modulation from block to block occurred at individual electrode sites. This is an indication of some volatility of the signal collection performance of the dry electrodes that was here treated by averaging power distributions across electrode sites.

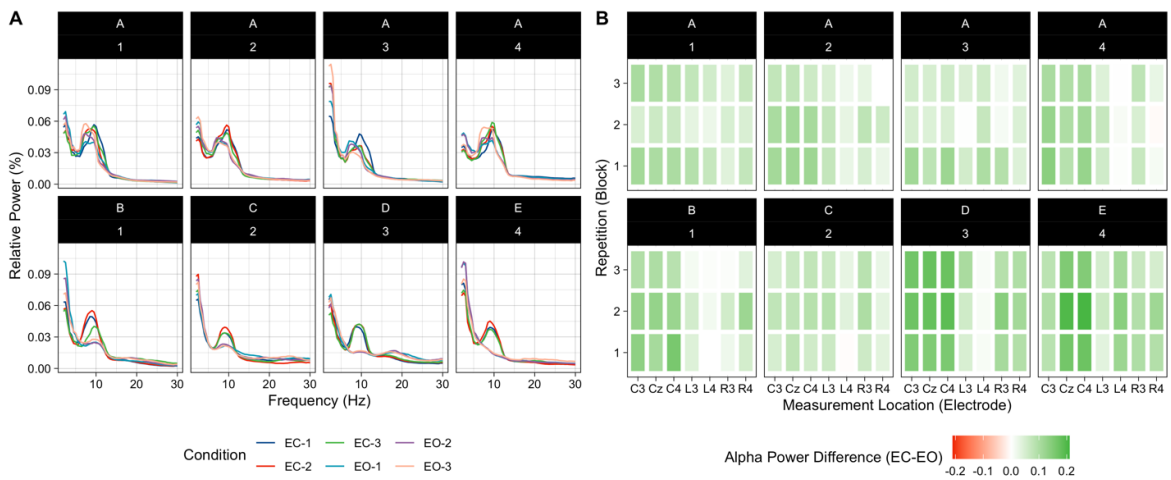

Figure 5. Berger effect replication – Alpha band power differences are shown for each participant. The top row shows the repeated single subject recordings. The bottom row shows four different participants. (A) shows the PSD in EC and EO phases, (B) the difference between EC-EO per electrode and block.

Next, the cognitive load modulation was assessed. For this purpose, the TLX reports were aggregated in the recommended form (i.e., by summing the six items) for each measurement occurrence. A linear mixed model (LMM) with random intercept for each session was used to assess a main effect of task on perceived load (including the eyes open rest and both mental arithmetic tasks). An ANOVA based on this model confirms a clear main effect ($p < .0001$) with significant post-hoc contrasts that indicate a stepwise increase from rest to easier to harder mental arithmetic tasks (all $p < .0001$) – see Figure 6.

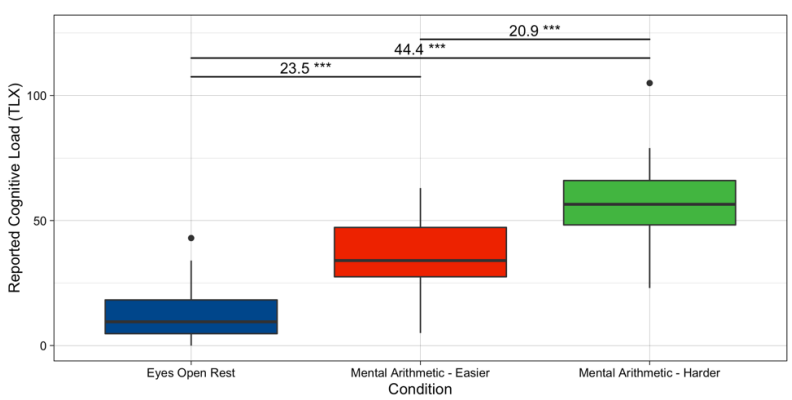

Figure 6. Load manipulation check study 1 – The perceived cognitive load increases from rest to easier to harder mental arithmetic. Values above boxes show median differences.

Given this confirmed manipulation, we pursued a direct investigation of the perceived cognitive load with Theta and Alpha band power changes. Repeated-measures (RM) correlations were computed to account for

the substantial inter-individual difference in EEG recordings (Blankertz et al., 2016). Figure 7 shows the strong and significant positive (Theta) and negative (Alpha) correlations between frequency band powers and cognitive load reports. Notably, the correlations are robust for all the recording groups (the single-subject – WS, and the multi-subject – BS recordings), and for most electrode sites. This finding is expected and in line with related work (see, e.g., Wascher et al., 2019). However, the results also highlight that Theta relationships are more pronounced on the top of the head and that Alpha power relationships are weaker in channels L3 & L4. After additional signal inspection, we believe that the latter effect is most likely attributed to a noisy channel in two sessions. As the sample was very small, these channels were not removed. The aggregation to "Top" and "Ears" regions of interest (ROI) appears to manage this limitation.

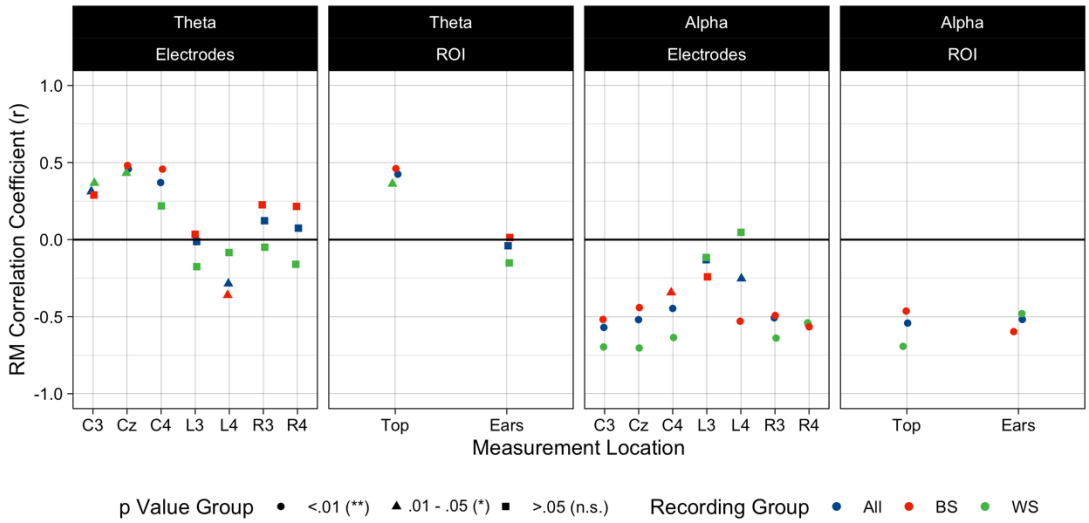

Figure 7. Correlations between EEG band powers and reported cognitive load.

While these results provided good initial support for the main research question of detecting cognitive load changes due to such strong effects, it was still possible, that the findings are limited to the restricted experiment and small sample. Therefore, we pursued an extended second study with a larger sample, an additional difficulty variation, and two additional tasks to further assess the findings' robustness.

Study 2

Procedure

For the second study, we investigated three main effects: (1) the reliability of the load-frequency band relationships across more difficulty levels, participants, and tasks, (2) the influence of possible confounding variables on this relationship, and (3) we explored the relationship between cognitive load and flow experiences – a mental state with strong ties to productivity and well-being that is increasingly investigated in the IS discipline and the context of knowledge work (Knierim et al., 2019; Riedl et al., 2020). We explore this relationship based on a central premise of flow theory, that flow can emerge under situations of optimal challenge (Engeser & Rheinberg, 2008) – or, as we will elaborate on below, optimal cognitive load.

A set of tasks was chosen that represents primarily cognitive tasks, yet with a variation in mental, sensory, and motoric demands. Specifically, we implemented an experiment that manipulates task difficulty in mental arithmetic, puzzle, and text copying tasks, with three difficulty levels (easy, medium, and hard). The difficulties for each task and condition were calibrated in a pre-test of the experiment with six participants.

In the (extended) mental arithmetic task, participants had up to 18 seconds to add a manipulated number of summands (2, 3, and 5 summands in the easy, medium, and hard task conditions respectively). The design was closely aligned with previous work (Knierim et al., 2019) and is otherwise identical to Study 1.

In the puzzle task, participants had up to 30 seconds to assemble a picture puzzle with a manipulated number of pieces (2, 6, and 12 pieces in the easy, medium, and hard task conditions respectively). This task

was added in contrast to the mental arithmetic task to provide a more visually stimulating task. Variations in visual stimulations have been documented as a possible challenge for cognitive load monitoring, especially as posterior Alpha band activity can be blocked by this factor (Blankertz et al., 2016). The task design was based on a related puzzle difficulty manipulation experiment (Tse et al., 2016).

In the text copying task, participants had up to 10 seconds to transcribe the text from a picture with a manipulated number of characters (9-10, 31-35, and 61-65 characters in the easy, medium, and hard task conditions respectively). This task was selected to provide additional variation in required motor action. Motoric activity (muscle movement) can represent a strong source of signal noise (Müller-Putz et al., 2015). Thereby, the robustness of the load-frequency relationships can be assessed further. The task design was closely related to a previous text copying difficulty manipulation experiment (Fest et al., 2021). Excerpts from a fiction novel were used as the content of the text snippets.

For all tasks, to ensure the differentiation of the easy and medium difficulty, the easy task condition showed only one of two pre-selected stimuli (e.g., only one of two puzzles – randomly drawn). Also, in all tasks, the task round lasted 90 seconds, with 4-second breaks between two consecutive stimuli. Examples of the difficulty in each task condition and the respective trial durations per task are shown in Figure 8. After an introductory task round in medium difficulty, participants completed a hard, then a medium, and lastly an easy difficulty task round. The order of the three tasks was randomized, with six possible orders. Each task round was followed by a short questionnaire including the NASA TLX to assess the perceived cognitive load, and three items from the flow short scale (FKS - Engeser & Rheinberg, 2008), with each item corresponding to the main flow dimensions in this scale (optimal challenge = "I felt just the right amount of challenge", fluency = "My thoughts/activities ran fluidly and smoothly", absorption = "I was totally absorbed in what I was doing"). We opted for this questionnaire design to limit the number of total questions, as the frequency of surveys was high. Lastly, each task type was preceded by an eyes-open resting stage.

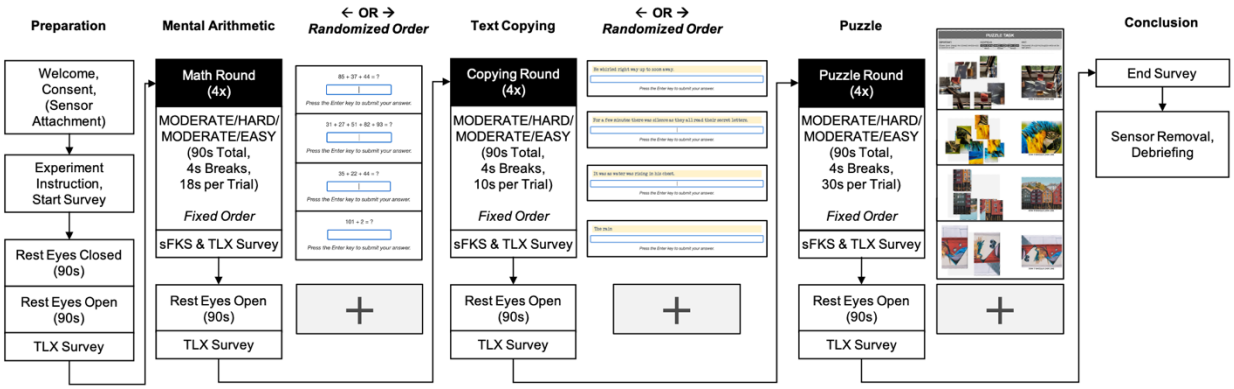

Figure 8. Study 2 procedure overview.

The data was again recorded with OpenBCI GUI and the experiment conducted through a browser-interface (oTree – Chen et al., 2016), again with local timestamp collection for precise synchronization of the EEG data and the experiment markers. The recruiting procedure and the experiment setup process were identical to study 1. The average duration of the experiment was 40.20 minutes. Overall 18 participants (6 female, age range = 21-34, median age = 27, 9 wearing glasses during the experiment) completed a single session of this experiment, thereby ensuring that each of the six possible task orders was completed three times. For one participant, only half of the experiment data was recorded due to a power outage in the experiment PC. Another participant was removed due to atypical EEG recordings, indicating that either the reference or ground electrode lost contact early during the experiment. Together with the absence of recording failures in study 1 the recording system thereby demonstrates good reliability. In this experiment, we also tracked the setup time from putting the headphones on to the start of the experiment to quantify some of the system's usability aspects. The system was always set up by the same experimentator. The median setup duration (including signal quality inspection) was 4.14 minutes (min = 1.11 minutes, max = 13.91 minutes). Wilcoxon rank test results indicate a significant difference in setup time for gender ($W = 60$, $p = .0245$ – possibly caused by longer hair), and no significant difference in setup duration for participants wearing glasses ($W = 49$, $p = .4894$).

Results

Following the recording, the EEG data were processed as outlined in the methods section above. The self-reports were aggregated using the recommended methods (sum for the TLX and mean for the FKS). Before aggregating the FKS, we inspected the internal consistency of our shortened scale version, which was deemed acceptable with a Cronbach's Alpha of .77.

As a manipulation check, we initially assessed the distribution of cognitive load (TLX) reports across the tasks. An LMM for each task with a random intercept for each session was used to assess main effects of difficulty conditions on perceived load. Ensuing ANOVA models confirm a clear main effect ($p < .0001$ for all three tasks) with significant post-hoc contrasts that indicate a stepwise increase from easier to medium, to harder difficulty conditions (all $p < .0001$) – see Figure 9. Notice, however, the clearer contrasts between easy and medium difficulties (when compared with the contrasts between medium and hard difficulties).

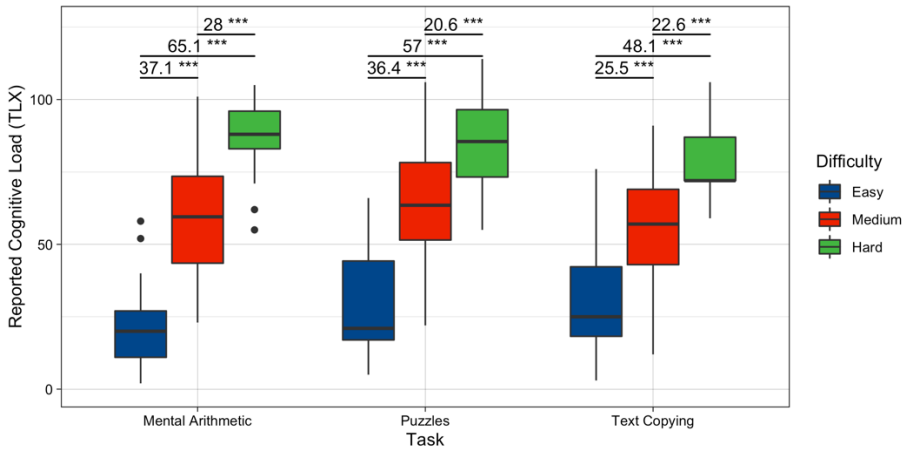

Figure 9. Load manipulation check study 2 –The perceived cognitive load increases from easier to harder task conditions. Values above boxes show median differences.

Afterwards, the relationships between TLX reports and Theta and Alpha frequency bands were assessed. For this purpose, LMMs with frequency band power as a fixed effect and participant as random intercept were fitted for the two main ROIs: mean average band power on the top of the head (C3, Cz, C4), and around the ear (L3, L4, R3, R4). The models' results are shown in Table 1. Significant positive relationships with Theta band power and highly significant negative relationships with Alpha band power are found in both regions. For the Theta band, the relationship with cognitive load is substantially weaker around the ears. The models also show a promising degree of explained variance, especially under consideration of both the random and the fixed effects.

Freq. Band	**Theta**		**Alpha**	
ROI	**Top**	**Ears**	**Top**	**Ears**
Power Estimate (p Value)	433.21 *** ($p < .0001$)	98.53 * ($p = .0199$)	-299.04 *** ($p < .0001$)	-277.93 *** ($p < .0001$)
R^2 (Marg. / Cond.)	.35 / .64	.03 / .06	.16 / .19	.15 / .19

Table 1: LMM results – EEG band power influence on reported cognitive load.

To further assess the possible influence of confounding variables – here: gender, glasses, and time, we included these variables as covariates in the aforementioned models. No significant relationships were found, which is why simple models were kept further throughout the remaining analyses. For a simplified relationship strength metric, please see additional RM correlation metrics in Figure 10. Here, it can also be seen that the relationship strength does not differ substantially for the three tasks.

Next, to assess the dependency of the relationship strength on experiment contrasts (i.e., to perform a sensitivity analysis), we systematically removed difficulty conditions from the data. For the first set (termed the "harder" set) we removed the easiest condition, for the second set (termed the "easier" set) the hardest condition, and for the third set all conditions except the two moderate difficulty rounds (termed the

"medium" set). The results in Table 2 show that the load-power relationships become weaker, especially with the removal of the easier conditions, and especially so for the Theta frequency band. The effect remains significant and strong for the Alpha band, despite the contrast removal. This indicates a higher sensitivity for the Alpha band for load detection given the current experiment data and feature extraction process.

Freq. Band	**Theta**		**Alpha**	
ROI	**Top**	**Ears**	**Top**	**Ears**
"Harder" Set (data without easiest conditions)				
Power Estimate (p Value)	98.45 * (p = .0281)	40.896 (p = .3270)	-193.23 ** (p = .0028)	-186.80 ** (p = .0057)
R^2 (Marg. / Cond.)	.05 / .13	.01 / .06	.06 / .07	.06 / .08
"Easier" Set (data without hardest conditions)				
Power Estimate (p Value)	236.71 *** (p = .0005)	126.01 ** (p = .0075)	-270.87 *** (p < .0001)	-227.50 *** (p < .0001)
R^2 (Marg. / Cond.)	.22 / .40	.07 / .12	.17 / .24	.14 / .19
"Medium" Set (data with only the medium difficulty conditions)				
Power Estimate (p Value)	26.84 (p = .5701)	11.62 (p = .8210)	-209.30 ** (p = .0091)	-191.50 * (p = .0155)
R^2 (Marg. / Cond.)	.01 / .31	.01 / .30	.12 / .26	.11 / .29

Table 2: Sensitivity analysis for EEG band power – reported cognitive load effects.

To further evaluate the relationship strength at a less aggregated level, Pearson correlations between TLX reports and frequency band powers were calculated for each individual participant. While the sample size for each individual is small, the average size of the correlation coefficients is in line with the previous analyses, showing strong load report relationships for Theta power on the top of the head and Alpha power in both ROI (see Figure 10). This plot also highlights that gender and glasses do not appear to systematically influence the strength of the load-power relationships. Interestingly, however, there is a substantial variance present for the load-frequency relationships at the individual level. On the one hand, this could be variance due to the small per-subject data points. On the other hand, it could indicate a potential for further improvement in the feature extraction approach.

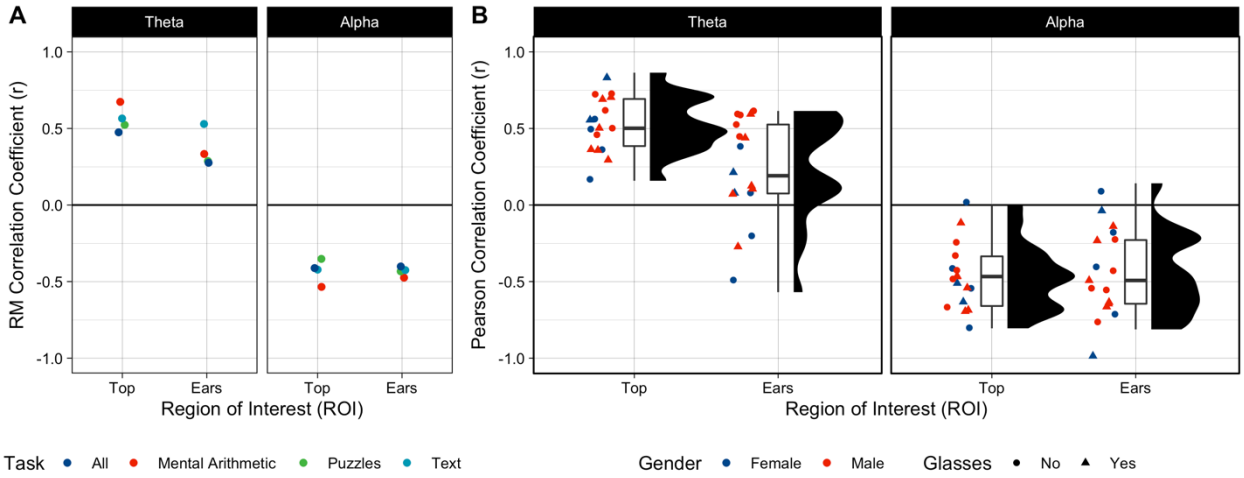

Figure 10. Correlation coefficients for reported cognitive load – EEG band power relationships across tasks (A) and for individual participants (B).

Given the reliable load-band power relationships, we deemed it reasonable to further explore the potential of extending the investigation towards the connection between load, frequency bands, and flow experience. A central premise in the underlying flow theory is that the so-called "peak experience" can emerge under situations of optimal challenge when a balance of an individual's abilities and the tasks' demands occurs (Engeser & Rheinberg, 2008). In extension, we propose, that such a situation is more likely present, with moderate levels of cognitive load. In contrast, underload or overload are more likely related to the situations

in which a task is too easy or too hard for an individual. Some previous research has proposed a related pattern between flow and mental effort (Ewing et al., 2016). Also, it is repeatedly documented, that flow experience shows an inverted U-shaped relationship with a task's difficulty (Engeser & Rheinberg, 2008; Knierim et al., 2019; Tozman et al., 2015). However, we are not aware of any research that has directly investigated the relationship between flow experience and subject (reported) and objective (frequency band) indicators of cognitive load. Therefore, to pursue this possibility for theoretical integration, we inspected the relationship between these variables. Using an LMM with load as a predictor for flow and a random intercept for participants, we initially found no significant, linear relationship. However, as previously mentioned, a non-linear relationship between the two variables is more likely to be expected. Adding a second order polynomial load term to the model shows a highly significant quadratic relationship ($p < .0001$ – see Figure 11). The substantial R^2 (marginal = .26, conditional =.42) further supports the strength of this relationship.

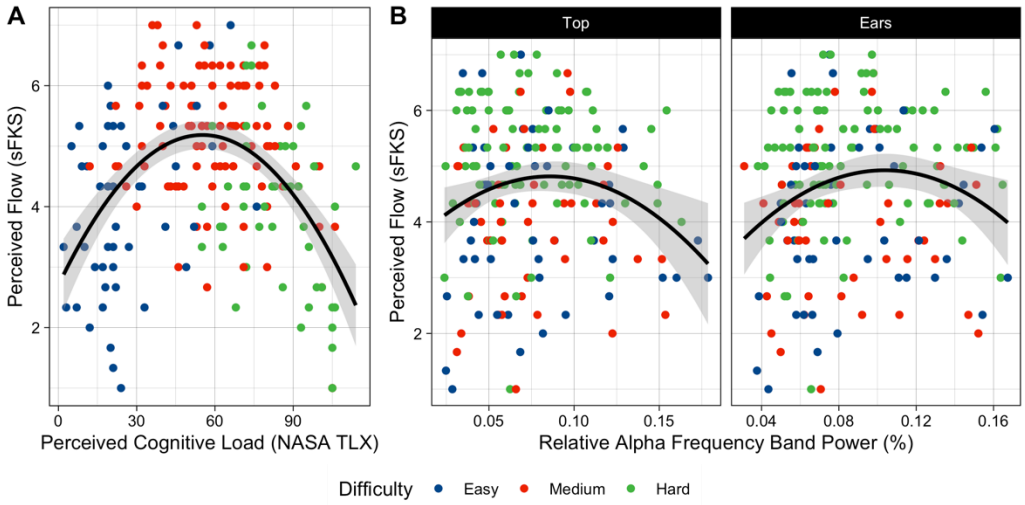

Figure 11. Inverted U-shape relationships between reported flow with reported cognitive load (A) and reported flow with Alpha band power (B).

Given the strength of the self-report relationship, we also explored the connection between the EEG load indicators and flow with linear and quadratic mixed models (see Table 3). For Theta band power, no significant relationships are found. In contrast, for Alpha band power significant quadratic relationships are found on the top of the head and in the ear region. The interpretation and implications of these findings are discussed in the following section.

Freq. Band	**Theta**		**Alpha**	
ROI	**Top**	**Ears**	**Top**	**Ears**
Linear Power Estimate (p Value)	-0.61 (p = .7952)	-2.01 (p = .4180)	-0.31 (p = .9370)	2.81 (p = .4690)
Quadr. Power Estimate (p Value)	-2.11 (p = 0.2630)	-0.87 (p = .6321)	-3.70 * (p = .0202)	-3.61 * (p = .0255)
R^2 (Marg. / Cond.)	0.01 / 0.17	0.01 / 0.15	0.04 / 0.17	0.04 / 0.15

Table 3: QMM results – EEG band power influence on flow.

Discussion

The presented work aims to evaluate the applicability of 3D-printed dry-electrode EEG headphones for cognitive load monitoring. Thereby, we investigate if the principle of EEG-based load monitoring can be leveraged in a more convenient and ubiquitous form factor. The results from two controlled experiments show promising support for this purpose, yet also highlight remaining limitations and challenges.

First of all, it should be acknowledged, that the identified positive Theta and negative Alpha power relationships with variations of cognitive load are highly in line with previous EEG research (Berka et al.,

2007; Blankertz et al., 2016), as has been recently emphasized by a meta-analysis on this matter (Chikhi et al., 2022). Thereby, we assume a high internal validity of the presented findings, that is also supported by the reliability of the observations across studies, sessions, and tasks. We want to emphasize that these observations were achieved even without the use of dedicated measures to improve signal qualities that are typically used in EEG research. These measures often include hair washing, skin preparation, bad channel treatment, and stationary artefact removal (e.g., for eye blink artefacts or similar patterns – see Blankertz et al., 2016; Müller-Putz et al., 2015). Thereby, our findings not only confer with this established knowledge in the direction of the power-load effects but also with their location as Theta power load modulation is typically observed over frontal midline areas and Alpha power modulation at posterior brain regions (Chikhi et al., 2022). This observation highlights the utility of including electrodes on the top of the head for Theta change observation, which is, for example, not a feature that is available in around-the-ear cEEGrid electrodes. Therefore, the presented headset provides some advantages for load monitoring over other, wearable and transparent EEG systems.

Yet, while Chikhi et al. (2022) report Theta changes to be the most robust with load modulation, our findings stand in contrast as the Alpha power changes appeared clearly more reliable, and, importantly, more sensitive to changes in cognitive load. We consider two main explanations and approaches to overcome this limitation. On the one hand, it is possible that signal artefacts or bad channels contaminated the Theta band more strongly than the Alpha band that is typically pronounced in adults during wakefulness (Müller-Putz et al., 2015 – see also Figure 5). This explanation also appears possible, as related work reports on the extensive signal-cleaning efforts that were required to extract reliable frequency band load estimators due to the presence of movement artefacts (Wascher et al., 2019). On the other hand, it has been documented repeatedly, that especially the precise extraction of Theta band power, can be a challenging endeavor, as it is not represented similarly saliently as the Alpha band in the EEG signal PSD (Ewing et al., 2016; Naumann et al., 2016; Vallarino et al., 2021). It is therefore possible, that our present signal processing approach is insufficient in terms of artefact and/or bad channel removal, and/or Theta frequency extraction. The substantial variation in per-subject correlations of frequency band activity and perceived load would lend support to this proposition. More sophisticated signal processing methods like artefact subspace reconstruction (ASR) for signal reconstruction in artefact-laden data epochs (see, e.g., Wascher et al., 2019), and more sophisticated spectral filtering methods for Theta frequency extraction have already been reported (Naumann et al., 2016). The latter work even reports the potential of eliciting very high Theta power load sensitivity, up to the point of being able to predict the exact difficulty level in a Tetris game. Future work on the EEG headphone system should investigate the use of such approaches, not only for the sake of methodological improvements but for practical reasons. Reliance on Alpha power alone might be short-lived for load prediction in real-world environments as the power in the Alpha band can quickly become blocked in situations with strongly changing visual demands (Blankertz et al., 2016).

Whether this is the case in knowledge work settings (especially such that include solitary work in front of a desktop PC), still needs to be investigated. The same needs to be said for other factors that could influence the systems capability for cognitive load monitoring, that have not yet been addressed systematically in this work. For example, while we do not find confounding influences of time on frequency band powers, such influences are known from neuroergonomics research, that investigates the detection of fatigue in system operators (for a discussion see Blankertz et al., 2016). Another factor is the question of how normal bodily movement during computerized work influences the usability of the Theta and Alpha bands for load detection. For these reasons, we propose to follow up our work with a study that bridges the controlled setting of the herein reported experiments with a controlled experience sampling study in the lab. In such a setting, study participants can complete controlled tasks (e.g., the mental arithmetic task in varied difficulty) and work on their own knowledge work project (e.g. a programming or scientific writing project) while being in a controlled environment. Data labels would then be generated by interrupting participants in a random interval (e.g., of 10, 15, or 20 minutes) during their typical work. Through multi-session participation, for example, for four one-hour sessions per day, a combination of internally valid and externally valid data can be recorded efficiently, while being able to monitor the correct administration of the sensing technology. Such a study design is proposed in Figure 12.

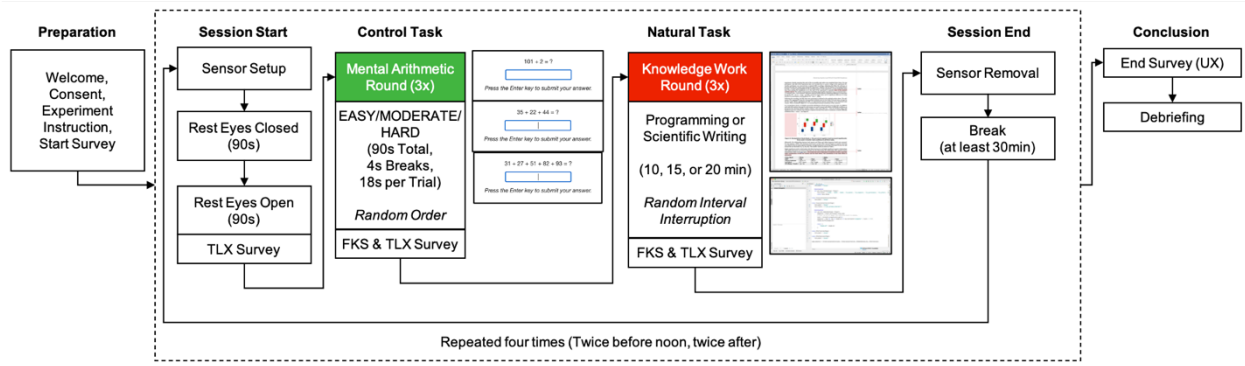

Figure 12. Follow-up study procedure overview.

Finally, beyond the principle load detection potentiality, the last findings of our studies indicate an interesting application direction for the EEG headphone system. While cognitive load is often considered as relevant to productivity and health by the avoidance of high load (Galy et al., 2012; Lyu et al., 2015), it is rarely discussed, which positive outcomes might be realized from optimizing load levels. With the AI-driven growth of the knowledge work population (Frey & Osborne, 2017), it becomes an important goal for workers and organizations to manage not only overload and stress, but also foster positive psychological experiences. For this reason, we explored the relationship between cognitive load and flow experience, a mental state that is positively associated with peak performance, growth and subjective well-being (Engeser & Rheinberg, 2008; Knierim et al., 2019). Because flow experiences are said to be likely when a task is optimally challenging, it appeared reasonable to draw a parallel to situations of moderate cognitive load. Our findings on the inverted U-shaped load-flow report relationship and the supported quadratic fit for Alpha power and flow reports provide evidence that flow experiences could be observed and eventually fostered by monitoring load with our headphone EEG system. As the flow relationships with band powers were weak, it needs to be investigated further, how extensive this potential is. On the one hand, a quadratic Theta band relationship would have similarly been expected. As the Theta band showed less reliability for cognitive load changes too though, it is possible that too much heterogeneity masks the relationship. On the other hand, it is possible, that the flow-frequency relationship is biased on the report side from the shortened scale that we used to keep the questionnaire effort low during study 2. Future work should thus use a more comprehensive flow scale (e.g. the 10-item version by Engeser & Rheinberg, 2008) to avoid this limitation. Also, it is likely, that additional data variance is needed to realize the flow detection potential. Situations with a more intense flow experience should be observed to ascertain the existence of the observed relationship. Nevertheless, previous work has already hinted at the possibility of flow detection (Knierim et al., 2021), which is why the current results are considered promising.

As our dry-EEG headphone system might allow for more continuous and unobtrusive data collection, this theoretical and practical extension of cognitive load monitoring can be pursued well – if the reliability and usability of the system can be established further. Furthermore, as previous research has also shown relationships between heart rate variability changes and flow (Rissler et al., 2020; Tozman et al., 2015), and as previous cEEGrid work has shown that an ECG is retrievable from around-the-ear EEG recordings (Bleichner & Debener, 2017), the EEG headphone system might be the perfect candidate to elaborate on this application potential. As a matter of fact, this potential would also benefit the identification of cognitive load monitoring altogether, as EEG and ECG signal features are considered complementary and have even been recommended for this purpose (Chikhi et al., 2022).

Conclusion

Through two controlled experiments, the present research demonstrates the promising potential of new, dry-electrode EEG headphones for monitoring cognitive load changes across multiple settings. Our research makes essential methodological, practical, and theoretical contributions to the IS discipline. In terms of methodology, we provide repeated and robust evidence that changes in cognitive load are well-observable in a system that is built on established measurement principles. Thereby, this EEG application leverages its sensing methods' properties of unobtrusive, automatic, and real-time cognitive load observation. In practical terms, we provide a low-cost and easily reproducible hardware system with

material and assembly instructions. Thus, any interested IS researcher can put this sensing system to use and even develop similar ear-centered sensing systems ("earables" – see Röddiger et al., 2022). Lastly, in terms of theoretical knowledge, we empirically integrate the variable of cognitive load with flow experiences and highlight not only the conceptual but also the direct connection between neural correlates of cognitive load with perceived flow intensities.

Overall, while the system is not without limitations, we find the reliability and sensitivity of negative Alpha power changes with increased load, the clear observation of positive and substantial Theta power modulation, and the possibility of inferring flow experience intensities from these band powers exciting for the realm of knowledge work. This is because these observations are made possible by a cost-effective system that provides operational reliability and ease of use, as is demonstrated by the solid results despite the absence of extensive participant and data processing procedures. With the open-source and configurable nature of the system, scholars, enthusiasts, and practitioners are enabled to further pursue these research directions on their own for the sake of larger data set creations. Such broadened neurophysiological data sets will be essential as the foundation for mental state detection based on machine learning classifiers (Rissler et al., 2020). Therefore, such an openly available technology can hopefully serve as a catalyst for the development of positive adaptive technologies that not only prevent bad load situations like stress but can actively help to foster positive subjective experiences like flow. Importantly, this research is targeted at fruitful grounds in the IS domain, not only because the EEG headphones represent a sensing system in an accepted form factor, but also as the EEG is at the moment among the most frequent neurophysiological sensing method applied in the IS domain (Müller-Putz et al., 2015; Riedl et al., 2020).

References

Athavipach, C., Pan-Ngum, S., & Israsena, P. (2019). A wearable in-ear EEG device for emotion monitoring. *Sensors*, *19*(18), 1–16.

Babadi, B., & Brown, E. N. (2014). A review of multitaper spectral analysis. *IEEE Transactions on Biomedical Engineering*, *61*(5), 1555–1564.

Barthélemy, Q., Mayaud, L., Ojeda, D., & Congedo, M. (2019). The Riemannian potato field: a tool for online signal quality index of EEG. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, *27*(2), 244–255.

Berger, H. (1929). Über das Elektroenkephalogramm des Menschen. *Archiv Für Psychiatrie Und Nervenkrankheiten*, *87*(1), 527–570.

Berka, C., Levendowski, D. J., Lumicao, M. N., Yau, A., Davis, G., Zivkovic, V. T., Tremoulet, P. D., & Craven, P. L. (2007). EEG Correlates of Task Engagement and Mental Workload in Vigilance, Learning, and Memory Tasks. *Aviation, Space, and Environmental Medicine*, *78*(5), 231–244.

Blankertz, B., Acqualagna, L., Dähne, S., Haufe, S., Schultze-Kraft, M., Sturm, I., Uščumlic, M., Wenzel, M. A., Curio, G., & Müller, K.-R. (2016). The Berlin Brain-Computer Interface: Progress Beyond Communication and Control. *Frontiers in Neuroscience*, *10*, 1–24.

Bleichner, M. G., & Debener, S. (2017). Concealed, unobtrusive ear-centered EEG acquisition: Ceegrids for transparent EEG. *Frontiers in Human Neuroscience*, *11*(April), 1–14.

Brünken, R., Plass, J. L., & Leutner, D. (2003). Direct measurement of cognitive load in multimedia learning. *Educational Psychologist*, *38*(1), 53–61.

Chen, D. L., Schonger, M., & Wickens, C. (2016). oTree-An open-source platform for laboratory, online, and field experiments. *Journal of Behavioral and Experimental Finance*, *9*, 88–97.

Chikhi, S., Matton, N., & Blanchet, S. (2022). EEG power spectral measures of cognitive workload: A meta-analysis. *Psychophysiology*, 1–24.

Debener, S., Emkes, R., De Vos, M., & Bleichner, M. (2015). Unobtrusive ambulatory EEG using a smartphone and flexible printed electrodes around the ear. *Scientific Reports*, *5*, 1–11.

Engeser, S., & Rheinberg, F. (2008). Flow, performance and moderators of challenge-skill balance. *Motivation and Emotion*, *32*(3), 158–172.

Ewing, K. C., Fairclough, S. H., & Gilleade, K. (2016). Evaluation of an Adaptive Game that Uses EEG Measures Validated during the Design Process as Inputs to a Biocybernetic Loop. *Frontiers in Human Neuroscience*, *10*, 1–13.

Fest, S., Kvaløy, O., Nieken, P., & Schöttner, A. (2021). How (not) to motivate online workers: Two controlled field experiments on leadership in the gig economy. *The Leadership Quarterly*, *32*(6), 16.

Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to

computerisation? *Technological Forecasting and Social Change*, *114*, 254–280.
Galy, E., Cariou, M., & Mélan, C. (2012). What is the relationship between mental workload factors and cognitive load types? *International Journal of Psychophysiology*, *83*(3), 269–275.
Grimes, M. G., & Valacich, Joseph, S. (2015). Mind Over Mouse: The Effect of Cognitive Load on Mouse Movement Behavior. *Proceedings of the 36th International Conference on Information Systems (ICIS)*, 1–13.
Hart, S. G., & Staveland, L. E. (1988). Development of NASA-TLX (Task Load Index): Results of empirical and theoretical research. *Advances in Psychology*, *52*(1), 139–183.
Hu, P. J., Hu, H., & Fang, X. (2017). Examining the Mediating Roles of Cognitive Load With a Website : a Field Quasi -Experiment. *MIS Quarterly: Management Information Systems*, *41*(3), 975–987.
Kartali, A., Janković, M. M., Gligorijević, I., Mijović, P., Mijović, B., & Leva, M. C. (2019). Real-time mental workload estimation using EEG. *International Symposium on Human Mental Workload: Models and Applications*, 20–34.
Knierim, M. T., Berger, C., & Reali, P. (2021). Open-Source Concealed EEG Data Collection for Brain-Computer-Interfaces: Neural Observation Through OpenBCI Amplifiers with Around-the-Ear cEEGrid Electrodes. *Brain-Computer Interfaces*, *9*(3), 161–179.
Knierim, M. T., Nadj, M., Li, M. X., & Weinhardt, C. (2019). Flow in knowledge work groups - Autonomy as a driver or digitally mediated communication as a limiting factor? *Proceedings of the 40th International Conference on Information Systems (ICIS)*, 1–17.
Lyu, Y., Luo, X., Zhou, J., Yu, C., Miao, C., Wang, T., Shi, Y., & Kameyama, K. (2015). Measuring PPG-Based Stress-Induced Vascular Response Index to Assess Cognitive Load and Stress. *Proceedings of the ACM CHI'15 Conference on Human Factors in Computing Systems*, 857–866.
Martin, L., Hauret, L., & Fuhrer, C. (2022). Digitally transformed home office impacts on job satisfaction, job stress and job productivity. COVID-19 findings. *PLoS ONE*, *17*(3), 1–23.
Müller-Putz, G. R., Riedl, R., & Wriessnegger, S. C. (2015). Electroencephalography (EEG) as a research tool in the information systems discipline: Foundations, measurement, and applications. *Communications of the Association for Information Systems*, *37*(1), 911–948.
Naumann, L., Schultze-Kraft, M., Dähne, S., & Blankertz, B. (2016). Prediction of Difficulty Levels in Video Games from Ongoing EEG. *International Workshop on Symbiotic Interaction*, 125–136.
Oostenveld, R., & Praamstra, P. (2001). The five percent electrode system for high-resolution EEG and ERP measurements. *Clinical Neurophysiology*, *112*(4), 713–719.
Paas, F., Renkl, A., & Sweller, J. (2003). Cognitive load theory and instructional design: Recent developments. *Educational Psychologist*, *38*(1), 1–4.
Rashid, U., Niazi, I. K., Signal, N., & Taylor, D. (2018). An EEG experimental study evaluating the performance of Texas instruments ADS1299. *Sensors*, *18*(11), 1–18.
Riedl, R., Fischer, T., Léger, P.-M., & Davis, F. D. (2020). A decade of NeuroIS research: progress, challenges, and future directions. *ACM SIGMIS Database: The DATABASE for Advances in Information Systems*, *51*(3), 13–54.
Riedl, R., Minas, R. K., Dennis, A. R., & Müller-Putz, G. R. (2020). Consumer-Grade EEG Instruments: Insights on the Measurement Quality Based on a Literature Review and Implications for NeuroIS Research. *Lecture Notes in Information Systems and Organisation*, *43*, 350–361.
Rissler, R., Nadj, M., Li, M. X., Loewe, N., Knierim, M. T., & Maedche, A. (2020). To Be or Not to Be in Flow at Work: Physiological Classification of Flow using Machine Learning. *IEEE Transactions on Affective Computing*, *Xxx*, 1–12.
Röddiger, T., Clarke, C., Breitling, P., Schneegans, T., Zhao, H., Gellersen, H., & Beigl, M. (2022). Sensing with Earables: A Systematic Literature Review and Taxonomy of Phenomena. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, *6*(3), 135–210.
Tozman, T., Magdas, E. S., MacDougall, H. G., & Vollmeyer, R. (2015). Understanding the psychophysiology of flow: A driving simulator experiment to investigate the relationship between flow and heart rate variability. *Computers in Human Behavior*, *52*, 408–418.
Tse, D. C. K., Fung, H. H., Nakamura, J., & Csikszentmihalyi, M. (2016). Teamwork and flow proneness mitigate the negative effect of excess challenge on flow. *Positive Psychology*, *13*(3), 284–289.
Vallarino, E., Sommariva, S., Arnaldi, D., Famà, F., Piana, M., & Nobili, F. (2021). Transfreq: a Python package for computing the theta-to-alpha transition frequency from resting state EEG. In *bioRxiv*.
Wascher, E., Arnau, S., Reiser, J. E., Rudinger, G., Karthaus, M., Rinkenauer, G., Dreger, F., & Getzmann, S. (2019). Evaluating Mental Load During Realistic Driving Simulations by Means of Round the Ear Electrodes. *Frontiers in Neuroscience*, *13*(September), 1–11.